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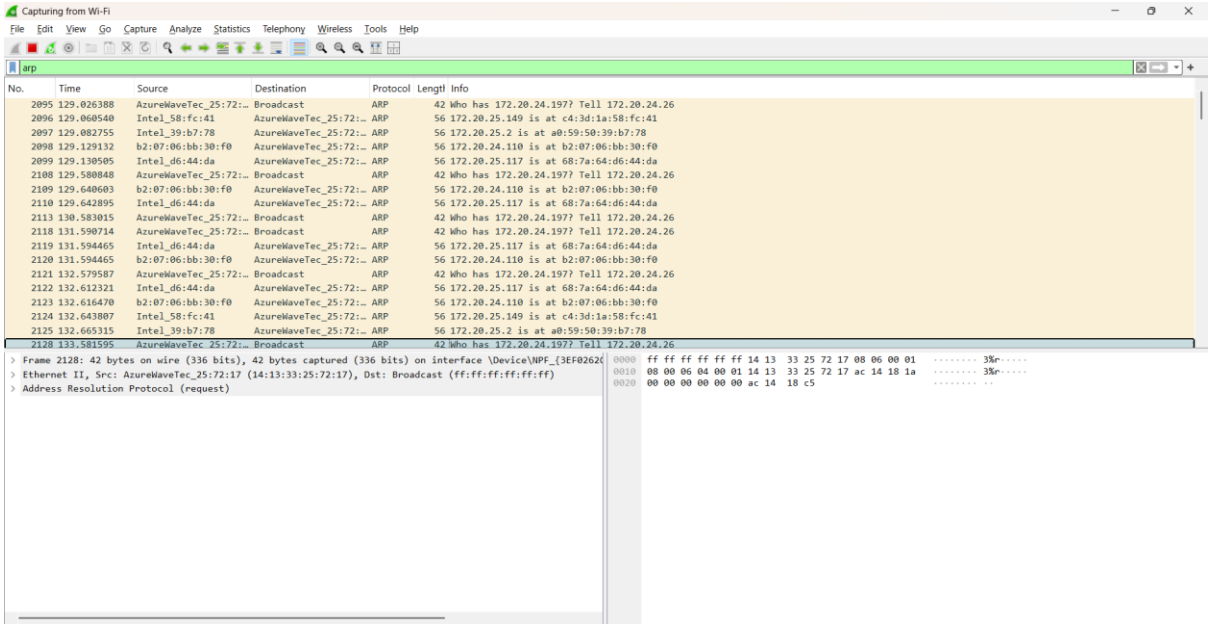
Question 1

Capture and analyze ARP packets using Wireshark. Inspect the ARP request and reply frames, and describe the role of the sender's IP and MAC address in these packets.

ARP (Address Resolution Protocol) is used to map an IP address to the corresponding MAC address in a local network. When a device wants to communicate with another device, it first sends an ARP request to find the MAC address associated with the destination IP address. The ARP request is broadcast to all devices in the network. The device that owns the requested IP address responds with an ARP reply containing its MAC address.

The sender's IP address represents the logical network address of the device initiating the ARP request, while the sender's MAC address represents the physical hardware address of that device. These values allow the receiving device to know where to send the ARP reply and help build the ARP table for future communication.

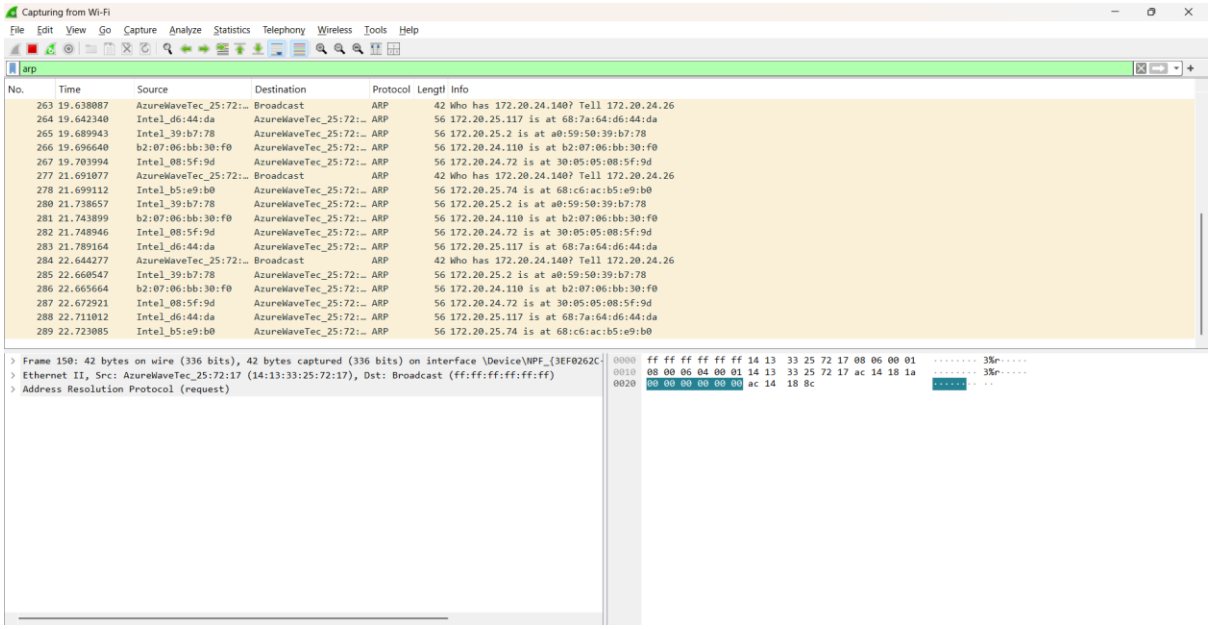
Pinging a device with IP address 172.20.24.197:



I have filtered out only the arp, so it will show only the fields like:

- Sender MAC address
- Sender IP address
- Target MAC address
- Target IP address

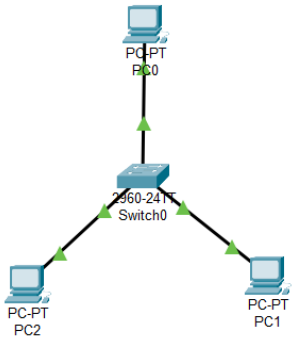
Pinging a domain like google with IP address 8.8.8.8:



Question 2

Using Packet Tracer, simulate an ARP spoofing attack. Analyse the behaviour of devices on the network when they receive a malicious ARP response.

The topology:



To demonstrate a spoofing attack, I am making the PC0 as the victim, PC1 as the gateway and the PC2 as the attacker.

PC0 (Victim)

- IP Address : 192.168.1.2
- Subnet Mask : 255.255.255.0
- Gateway : 192.168.1.1

PC1 (Gateway)

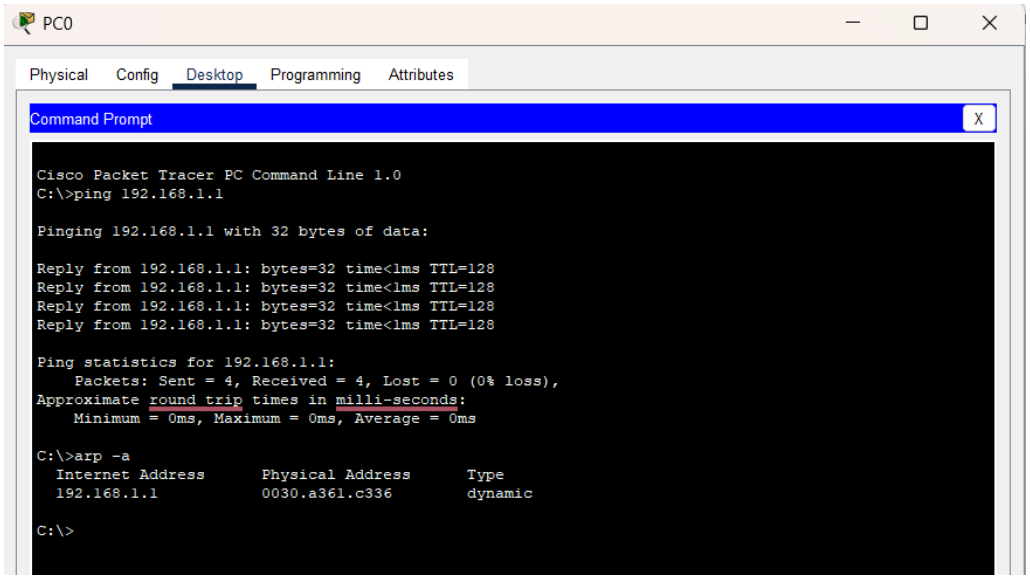
- IP Address : 192.168.1.1
- Subnet Mask : 255.255.255.0

PC2 (Attacker)

- IP Address : 192.168.1.3
- Subnet Mask : 255.255.255.0

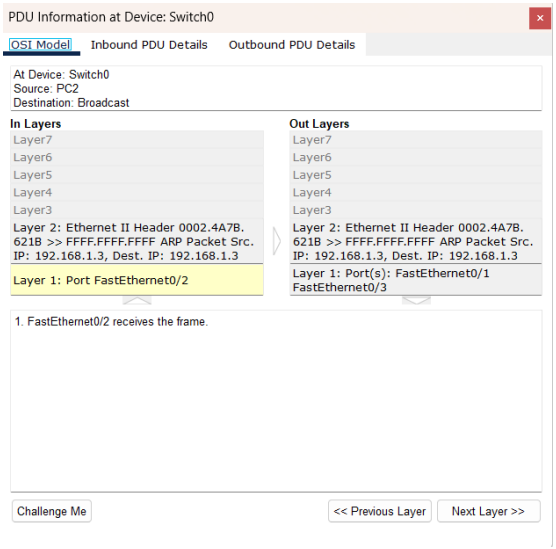
The normal behaviour without any spoofing:

Pinging PC1 from PC0 and checking the arp table:



The spoofing (The attacker sends a fake ARP reply):

After pinging the gateway from PC0:



In the Packet Tracer simulation, the ARP request packet was observed using the OSI Model view. The ARP request was broadcast from PC2 with the destination MAC address FFFF.FFFF.FFFF, which indicates that the packet is sent to all devices in the network. The switch receives the frame and forwards it to all connected ports except the incoming port. This allows the device with the matching IP address to respond with an ARP reply containing its MAC address. In an ARP spoofing attack, a malicious device can send a forged ARP reply claiming ownership of another device's IP address. If the victim accepts this reply, it updates its ARP table incorrectly, causing traffic to be redirected through the attacker.

Simulation Procedure

A small network was created in Cisco Packet Tracer consisting of three PCs connected through a switch. IP addresses were assigned to each device within the same subnet. Packet Tracer was switched to Simulation Mode and the ARP protocol filter was enabled. When a ping command was executed from one PC to another, the ARP request and ARP reply packets were observed in the simulation.

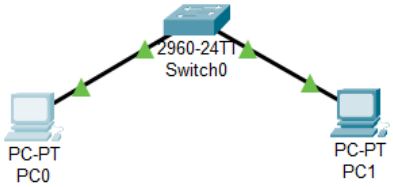
Analysis

The ARP request is broadcast to all devices in the network asking for the MAC address associated with a specific IP address. The device that owns the requested IP address responds with an ARP reply containing its MAC address. In an ARP spoofing attack, a malicious device sends a fake ARP reply claiming that its MAC address corresponds to the IP address of another device, such as the gateway. This causes the victim device to update its ARP table incorrectly, allowing the attacker to intercept network traffic.

Question 3

Manually configure static IPs on the client device (like PC or your mobile phone) and verify connectivity using ping.

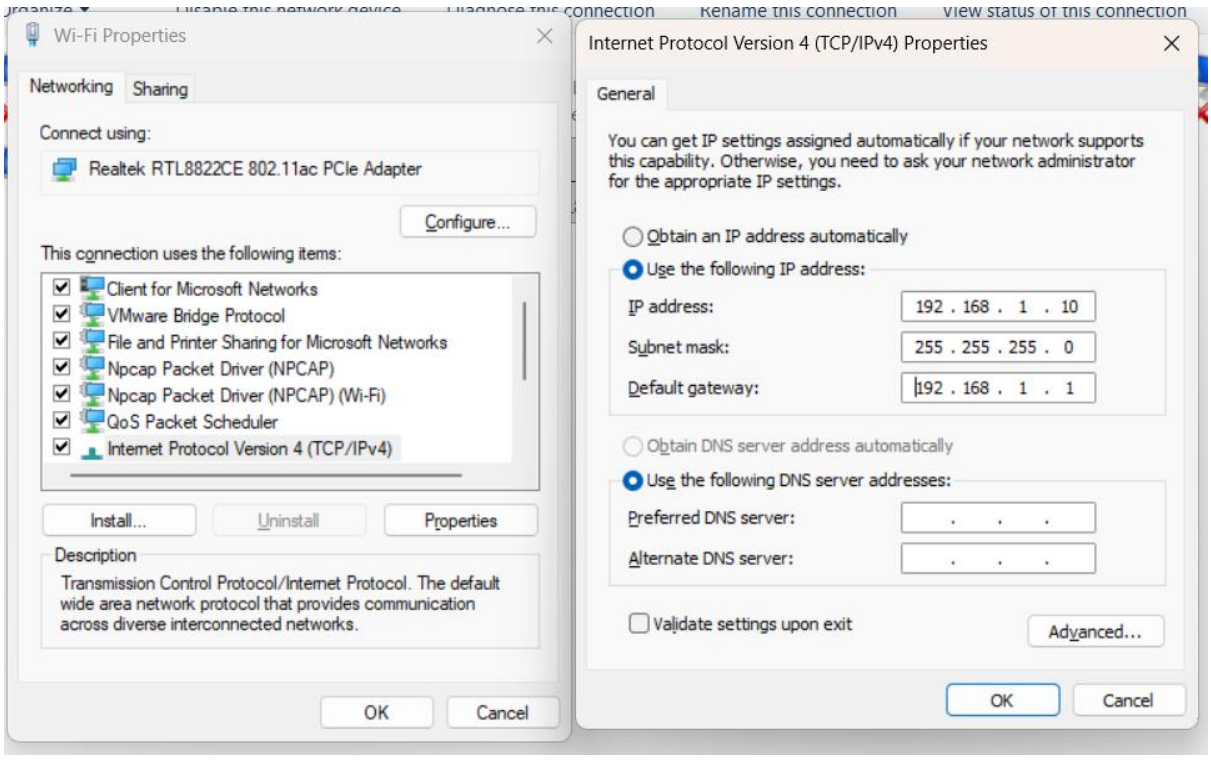
Using Cisco Packet Tracer:




```
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

In this task, static IP addresses were manually configured on the client devices using the IP configuration settings in Cisco Packet Tracer. PC0 was assigned the IP address 192.168.1.2 and PC1 was assigned the IP address 192.168.1.3 with a subnet mask of 255.255.255.0. After configuring the addresses, connectivity between the devices was verified using the ping command. The successful ping replies confirmed that the devices were able to communicate with each other within the same network.

Real-Time Static IP Configuration:



I have set the static IP address manually

```
C:\Users\91866>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 122.184.65.193: Destination net unreachable.
Reply from 122.184.65.193: Destination net unreachable.
Reply from 122.184.65.193: Destination net unreachable.
Reply from 122.184.65.193: Destination net unreachable.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\91866>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 172.20.24.1: Destination net unreachable.
Reply from 172.20.24.1: Destination net unreachable.
Reply from 172.20.24.1: Destination net unreachable.
Reply from 172.20.24.1: Destination net unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Have pinged both the google domain and a PC.

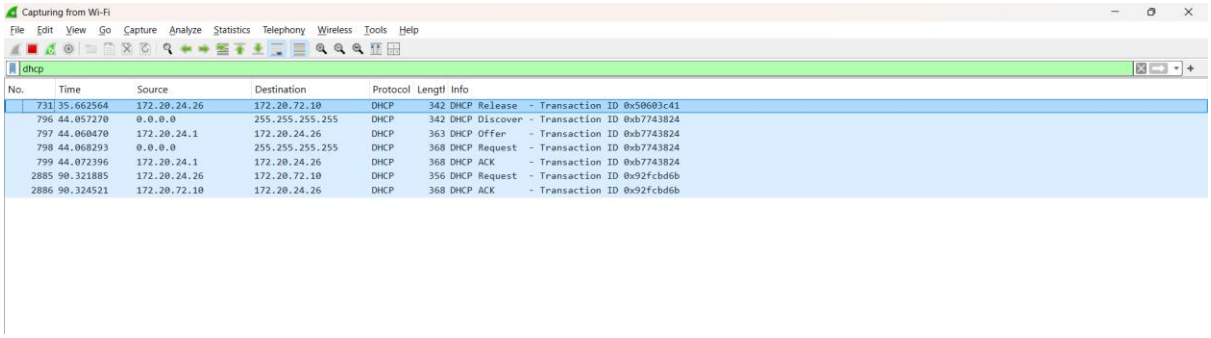
In this task, a static IP address was manually configured on the client device using the IPv4 configuration settings in the operating system. The IP address, subnet mask, and default gateway were entered manually. After configuring the static IP address, the ipconfig command was used to verify the configuration. Connectivity was then tested using the ping command to check communication with another device in the same network. The successful ping replies confirmed that the static IP configuration was correct and that the devices were able to communicate within the network.

Question 4

Use Wireshark to capture DHCP Discover, Offer, Request, and Acknowledge messages and explain the process.

This process is called **DORA**:

- Discover
- Offer
- Request
- Acknowledge



• **DHCP Discover:** The client broadcasts a request to locate available DHCP servers.
• **DHCP Offer:** The DHCP server responds with an available IP address and configuration details.
• **DHCP Request:** The client requests the offered IP address from the server.
• **DHCP Acknowledgement:** The server confirms the IP assignment and the client begins using the address.

In this experiment, Wireshark was used to capture DHCP communication between a client device and a DHCP server. Packet capture was started and the ipconfig /release and ipconfig /renew commands were executed to force the DHCP process. The captured packets showed the DHCP DORA sequence: Discover, Offer, Request, and Acknowledge. The client first broadcasted a DHCP Discover message to locate a DHCP server. The server responded with a DHCP Offer containing an available IP address. The client then sent a DHCP Request message to accept the offered address, and finally the server responded with a DHCP Acknowledgement confirming the assignment of the IP configuration parameters.

Question 5

Given an IP address range of 192.168.1.0/24, divide the network into 4 subnets.

Task:

- **Manually calculate the new subnet mask**
- **Find the range of valid IP addresses for each subnet**
- **Assign IP addresses from these subnets to devices in Cisco Packet Tracer**
- **Verify connectivity using ping**

Manual Calculation:

5. The given IP address : 192.168.1.0/24

We need to divide this network to 4 subnets

Total no of address

$2^{32-24} = 2^{8} = 256$ address

$\Rightarrow \frac{256}{4} = 64 = 2^{6}$

∴ The CIDR number is /26 (32 − 26 = 6)

Now:

The new subnet mask: 255.255.255.192

Subnet-1

Network address: 192.168.1.0

First valid host : 192.168.1.1

Last valid host : 192.168.1.62

Broadcast address: 192.168.1.63

Subnet-2

Network address: 192.168.1.64

First valid host : 192.168.1.65

Last valid host: 192.168.1.126

Broadcast address : 192.168.1.127

Subnet-3

Network address : 192.168.1.128

First valid host : 192.168.1.129

Last valid host : 192.168.1.190

Broadcast address : 192.168.1.191

Subnet-4

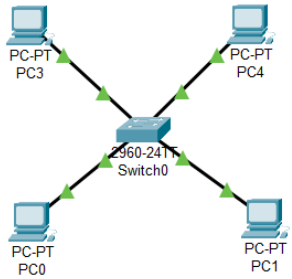
Network address : 192.168.1.192

First valid host : 192.168.1.193

Last valid host : 192.168.1.254

Broadcast address : 192.168.1.255

Packet Tracer Configuration:



Device	IP Address	Subnet
PC0	192.168.1.2	Subnet 1
PC1	192.168.1.66	Subnet 2
PC2	192.168.1.130	Subnet 3
PC3	192.168.1.194	Subnet 4

The ping from PC0 to PC1:

```
C:\>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Reply from 192.168.1.66: bytes=32 time<1ms TTL=128
Reply from 192.168.1.66: bytes=32 time<1ms TTL=128
Reply from 192.168.1.66: bytes=32 time<1ms TTL=128
Reply from 192.168.1.66: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

The ping from PC0 to PC3:

```
C:\>ping 192.168.1.194

Pinging 192.168.1.194 with 32 bytes of data:

Reply from 192.168.1.194: bytes=32 time<1ms TTL=128
Reply from 192.168.1.194: bytes=32 time<1ms TTL=128
Reply from 192.168.1.194: bytes=32 time=13ms TTL=128
Reply from 192.168.1.194: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

The ping from PC1 to PC0:

```
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The ping from PC2 to PC3:

```
C:\>ping 192.168.1.194

Pinging 192.168.1.194 with 32 bytes of data:

Reply from 192.168.1.194: bytes=32 time<1ms TTL=128
Reply from 192.168.1.194: bytes=32 time=1ms TTL=128
Reply from 192.168.1.194: bytes=32 time<1ms TTL=128
Reply from 192.168.1.194: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.194:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

The ping from PC1 to PC2:

```
C:\>ping 192.168.1.130

Pinging 192.168.1.130 with 32 bytes of data:

Reply from 192.168.1.130: bytes=32 time=1ms TTL=128
Reply from 192.168.1.130: bytes=32 time<1ms TTL=128
Reply from 192.168.1.130: bytes=32 time<1ms TTL=128
Reply from 192.168.1.130: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Four subnets were created from the network 192.168.1.0/24. For verification, devices were assigned addresses from the same subnet to test connectivity using the ping command.

Question 6

You are given three IP addresses: 10.1.1.1, 172.16.5.10, and 192.168.1.5.

Task:

1. **Identify the class of each IP address (Class A, B, or C).**
2. **Determine the default subnet mask for each class.**
3. **Provide the range of IP addresses for each class.**

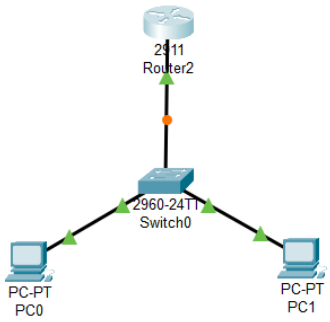
IP address	Address Range	Default Subnet Mask	Class
10.1.1.1	1.0.0.0 – 126.255.255.255	255.0.0.0	Class A
172.16.5.10	128.0.0.0 – 191.255.255.255	255.255.0.0	Class B
192.168.1.5	192.0.0.0 – 223.255.255.255	255.255.255.0	Class C

The class of an IP address is determined based on the value of its first octet. The IP address 10.1.1.1 belongs to Class A because its first octet falls within the range of 1 to 126, and its default subnet mask is 255.0.0.0. The IP address 172.16.5.10 belongs to Class B, as the first octet lies between 128 and 191, and its default subnet mask is 255.255.0.0. The IP address 192.168.1.5 belongs to Class C, since its first octet lies within the range of 192 to 223, and the default subnet mask for this class is 255.255.255.0.

Question 7

In Cisco Packet Tracer, create a small network with multiple devices (e.g., 2 PCs and a router). Use private IP addresses (e.g., 192.168.1.x) on the PCs and configure the router to perform NAT to allow the PCs to access the internet.

The topology:



Configure LAN interface:

```
Router#enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#
```

Configure WAN interface:

```
Router(config)#interface g0/1
Router(config-if)#ip address 200.1.1.1 255.255.255.0
Router(config-if)#ip nat inside
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```

To configure the NAT:

Allow the private network:

access-list 1 permit 192.168.1.0 0.0.0.255

Enable NAT overload:

ip nat inside source list 1 interface g0/1 overload

To verify the NAT Table:

The command on the router:

show ip nat translations

Output:

Inside global	Inside local
200.1.1.1	192.168.1.2

In this experiment, a small network consisting of two PCs, a switch, and a router was created in Cisco Packet Tracer. The PCs were assigned private IP addresses in the range 192.168.1.x. The router was configured to perform Network Address Translation (NAT) so that the private network could communicate with external networks. NAT was configured using an access list and NAT overload on the router's external interface. When the PCs attempted to ping an external IP address, the router translated the private IP address to its public IP address. This allowed communication between the private network and external networks.

Source IP Before and After NAT:

Stage	Source IP Address
Before NAT	192.168.1.2 (Private IP of PC)
After NAT	200.1.1.1 (Public IP of Router)